



COMPETENCY STANDARD
FOR
BUILDING ELECTRIFICATION
(CONSTRUCTION SECTOR)

Skills for Industry Competitiveness and Innovation Program (SICIP)
Finance Division, Ministry of Finance

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The Competency Standards for **Building Electrification** is a document for developing curricula, teaching and learning materials, and assessment tools. It also serves as the document for providing training consistent with the requirements of the industry for individuals who pass through the set standard via assessment. Subsequently, they would be qualified and settled for a relevant job.

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INTRODUCTION:

The Skills for Industry Competitiveness and Innovation Program (SICIP) has the overall objective of developing a skilled workforce adept at handling new technologies, especially for emerging industries in Bangladesh. It will expand skills training and strengthen the development of the training ecosystem to address the skills requirements of the SICIP-selected industry sectors. The program aims to (i) increase the technology-oriented skilled workforce across emerging and priority sectors, (ii) promote inclusive skilling and upskilling opportunities for women and socially disadvantaged groups, (iii) incentivize industry-university partnerships to nurture innovation capacity and improve industry competitiveness, and (iv) foster skills for climate-resilient manufacturing processes and green technologies. The program is expected to benefit about 220,000 new and existing workers over a 6-year implementation period from 2024-2029.

The SICIP Program has, therefore, taken the initiative to enhance the employability and productivity of trainees by implementing market-responsive and job-focused training programs through public and private training providers. This will require the development of competency standards for each of the occupations/trades which will provide a structured framework in the learning process to guide training providers, ensure consistent training quality, and create an alignment between the skills provided by the training institutes and the needs of the industry.

This competency standard is therefore developed to improve skills following the job roles and skill sets of the occupation and ensure that the required skills are aligned with industry requirements.

The Competency Standard also suggests integration of YouTube or similar digital platforms or downloaded clips into classroom practice to ensure simulated creation of the contents so that learners are exposed to visual demonstrations before classroom instruction or practical session, which aligns with modern learning preference and supports flipped classroom models.

The document details the format, sequencing, wording, and layout of the Competency Standard for an occupation which comprises Units of Competence and its corresponding Elements.

OVERVIEW:

A **Competency Standard** is a written specification of the knowledge, skills, and attitudes required for the performance of a job or occupation or trade corresponding to the standard of performance required in the workplace.

Competency standard:

- provides a consistent and reliable set of components for training, recognizing, and assessing people's skills, and may also have optional support materials.

- enables industry-recognized qualifications to be awarded through direct assessment of workplace competencies
- encourages the development and delivery of flexible training that suits individual and industry requirements
- encourages learning and assessment in a work-related environment, which leads to verifiable workplace outcomes.

Competency Standard has been developed by a working group comprised of occupation-specific experts from the industry/institution and relevant consultants of SICIP.

Competency Standards describe the skills, knowledge, and attitude needed to perform effectively in the workplace. Competency Standards acknowledge that people can achieve vocational and technical competency in many ways by emphasizing what the learner can do, not how or where they learned to do it.

With Competency Standards, assessment and training may be conducted at the workplace, at training organization, during regular work, or through work experience, work placement, work simulation or any combination of these.

A Unit of Competency describes a distinct work activity that would normally be undertaken by one person in accordance with industry standards.

Units of Competency are documented in a standard format that comprises:

- Reference to Industry Sector, Occupational Title and Occupational Description
- Unit code
- Unit title
- Unit descriptor
- Unit of Competency
- Elements and performance criteria
- Variables and range statement
- Evidence guides

Together all the parts of a Unit of Competency:

- Describe a work activity
- Guide the assessor in determining whether the candidate is competent.

The ensuing sections of this document comprise a description of the respective occupation with all the key components of a Unit of Competence:

- An overview of all Units of Competence for the occupation and their corresponding duration required for completion of training.
- The Competency Standards that include the Unit of Competency, Unit Descriptor, Elements and Performance Criteria, Range of Variables, Curricular Content Guide, and Assessment Evidence Guide.

Units & Elements at a glance:

OCCUPATION SPECIFIC COMPETENCIES (120HRS.)

Code	Unit of Competency	Elements of Competency	Duration (hours)
SICIP-LE-BE-01-O	Understand electrical system.	1. Identify types of tool/equipment and symbols in electrical system. 2. Identify fittings used in electrical works. 3. Measure Current, Voltage and Resistance. 4. Interpret earthing, changeover switch, magnetic contact and meters. 5. Identify sizes and capacity of wires and cables. 6. Maintain workplace, tools, equipment and materials.	15
SICIP-LE-BE-02-O	Perform installation of electrical circuit.	1. Prepare for works. 2. Install one lamp control by SPST and SPDT switch. 3. Install one calling bell from two points. 4. Prepare the circuit of LED Lamp. 5. Install ceiling fan with regulator. 6. Install power socket.	20
SICIP-LE-BE-03-O	Perform wire or cable termination.	1. Collect tools, equipment and materials. 2. Identify cable Lugs. 3. Connect cables (1-4 mm²). 4. Maintain workplace, tools and materials	5
SICIP-LE-BE-04-O	Perform cable glanding.	1. Collect tools, equipment and materials. 2. Identify cable Gland. 3. Connect plastic gland (PG-Gland).	5
SICIP-LE-BE-05-O	Perform channel wiring.	1. Prepare for works. 2. Carry out channel wiring.	35

Code	Unit of Competency	Elements of Competency	Duration (hours)
		3. Install boards and other accessories. 4. Test the wiring. 5. Maintain and store tools and materials.	
SICIP-LE-BE-06-O	Perform conduit wiring	1. Collect tools, equipment and materials. 2. Install conduits and set cables. 3. Install boards and other accessories of wiring. 4. Test the wiring. 5. Maintain workplace, tools and materials.	40
Total Hours			120

COMPETENCY STANDARD: BUILDING ELECTRICIAN

Occupation-specific Competencies

Unit of Competency: UNDERSTAND ELECTRICAL SYSTEM	Nominal Duration: 15 hrs.	Unit Code: SICIP-LE-BE-01-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to understand electrical system. It specifically includes the tasks of identifying types of tool/equipment and symbols in electrical system, identifying fittings used in electrical works, measuring current, voltage and resistance, interpreting earthing, change over switch, magnetic contact and meters, identifying sizes and capacity of wires and cables and maintaining workplace, tools, equipment and materials.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify types of tool/equipment and symbols in electrical system.	1.1 <u>Personal Protective Equipment (PPE)</u> is used while working. 1.2 <u>Electrical properties/parameters</u> are comprehended and measured using appropriate measuring tool/instrument. 1.3 Safety requirements are identified. 1.4 Hazards and Risk are explained. 1.5 <u>Tools, equipment</u> and symbols are comprehended required for electrical works. 1.6 Different types of tools and equipment are explained. 1.7 Electrical drawing is explained.
2. Identify fittings used in electrical works.	2.1 <u>Fittings</u> are comprehended required for electrical works. 2.2 Fittings are identified. 2.3 Function and properties of fittings are described.
3. Measure current, voltage and resistance.	3.1 Range of ammeter is set as per job requirement. 3.2 The ampere meter is connected in series with load as per circuit diagram. 3.3 Phase cable is inserted into the ring of the clip-on meter. 3.4 Range of voltmeter is set as per job requirement. 3.5 Voltmeter is connected in parallel with the load as per

	circuit diagram. 3.6 Voltmeter is kept in AC voltage range as required. 3.7 Ohm meter is connected in parallel with the load as per circuit diagram. 3.8 Meter pointer is observed and recorded as per workplace standard.
4. Interpret earthing, change over switch, magnetic contact and meters.	4.1 Types and <u>methods of earthing</u> are identified in accordance with electrical plan/design. 4.2 Types and sizes of <u>earthing materials</u> are identified in accordance with electrical plan/design. 4.3 Change over switch and magnetic contract is identified. 4.4 Types and functions of energy meters are identified.
5. Identify sizes and capacity of wires and cables.	5.1 Size of the cable is compared with the cable specification as per specific load. 5.2 Proper diameter/gauge of conductor is measured using SWG/micro meter. 5.3 Cable sizes are compared with capacity of wire and cables.
6. Maintain workplace, tools, equipment and materials.	6.1 Work area is cleaned in accordance with workplace procedures. 6.2 Unused materials are stored for re-use or disposed following workplace procedures. 6.3 Waste and scrap materials are disposed with following workplace and environmental procedures. 6.4 Inventory of tools equipment are conducted and recorded as per checklist. 6.5 Tools and equipment are cleaned and stored as per manufacturer's recommendation in appropriate location.

Range of Variables

Variable	Range May Include but not limited to:
1. Personal Protective Equipment (PPE)	1.1 Apron 1.2 Gloves 1.3 Safety shoes 1.4 Helmet 1.5 Face mask 1.6 Goggles and safety glasses

	1.7 Ear plug
2. Electrical properties/parameters	2.1 Voltage 2.2 Resistance 2.3 Current 2.4 Capacitance 2.5 Resistance
3. Tools and equipment	3.1 Hammer 3.2 Cutting pliers 3.3 Combination pliers 3.4 Wire stripper 3.5 Neon tester 3.6 Screw driver (flat, star and connecting) 3.7 Electrician knife 3.8 Hack saw blade 3.9 Tri-Square 3.10 Multi meter (Analog, Digital) 3.11 Series lamp 3.12 Insulation resistance tester (Megger) 3.13 Nose pliers 3.14 Hacksaw 3.15 Measuring tape 3.16 Drill Machine 3.17 Earth tester 3.18 Change over switch 3.19 Spirit level 3.20 wire stripper
4. Fittings	4.1 Switches 4.1.1 One-way switches 4.1.2 Two-way switches 4.1.3 Fan regulator switches 4.1.4 Push-button switches 4.2 Sockets 4.2.1 3-pin 5/6 Amp socket outlets 4.2.2 6/16 Amp universal sockets 4.3 Lighting Fixtures 4.3.1 Ceiling lamp holders 4.3.2 Wall lamp holders 4.3.3 Pendant lamp holders 4.3.4 LED bulbs and fittings 4.3.5 Fluorescent tube holders 4.4 Conduits 4.4.1 PVC conduits 4.4.2 Conduit bends (90°, 45° elbows) 4.4.3 Conduit clips/clamps

	4.4.4 Junction boxes 4.4.5 Back boxes 4.5 Connectors 4.5.1 Wire nuts / twist connectors 4.5.2 Terminal blocks 4.5.3 Cable lugs 4.6 Protective Devices 4.6.1 Miniature Circuit Breakers (MCBs) 4.6.2 Residual Current Circuit Breakers (RCCBs) 4.6.3 Fuses and fuse holders 4.7 Earthing 4.7.1 Earthing rods and clamps 4.7.2 Earthing wires and terminals
5. Methods of earthing	5.1 Pipe earthing 5.2 Rod earthing 5.3 Plate earthing 5.4 Waterman earthing 5.5 Strip or wire earthing
6. Earthing materials	6.1 Continuity conductor/cable 6.2 Earthing lead 6.3 Earth electrode / plate 6.4 Connector 6.5 G.I. pipe / conduit 6.6 Bolts and nuts 6.7 Powdered charcoal 6.8 Salt
7. Cable sizes	7.1 According to the chart

Curricular Content Guide:

1. Underpinning Knowledge	1.1 Define Electrical properties/parameters. 1.2 Types of tool/equipment and symbols in electrical system. 1.3 Describe electrical drawing. 1.4 Function and properties of fittings. 1.5 Define measure current, voltage and resistance. 1.6 Know the purpose and different types of earthing systems. 1.7 Types and functions of energy meters and change over switch. 1.8 Understand that wires and cables, importance of using the correct size and used in homes and buildings.
2. Underpinning Skills	2.1 Listing tools and equipment required for electrical works.

	2.2 Identifying fittings. 2.3 Measuring current, voltage and resistance 2.4 Identifying earthing points and test earthing resistance 2.5 Choosing and installing the correct size wire
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 1.1 Workplace (simulated or actual) 1.2 Various kinds of electrical/electronic components, tools and equipment 1.3 Papers

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 comprehended electrical properties/parameters. 1.2 measured electrical parameters using appropriate measuring tool/instrument. 1.3 identified types and methods of earthing. 1.4 compared cable sizes with capacity of wire and cables. 1.5 stored unused materials for re-use
2. Methods of Assessment	Methods of assessment may include but not limited to: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done at a training center. 3.2 Assessment should be done by the certified assessor of TÜV SÜD.

Unit of Competency: PERFORM INSTALLATION OF	Nominal Duration: 20 hrs.	Unit Code: SICIP-LE-BE-02-O
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ELECTRICAL CIRCUIT		
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to perform installation of electrical circuit. It specifically includes the tasks of preparing for works, installing one lamp control by SPST and SPDT switch, installing one calling bell control from two point, preparing the circuit of LED Lamp, installing ceiling fan with regulator and installing power socket.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Prepare for works.	1.1 Personal Protective Equipment (PPE) is collected and worn as per job requirement. 1.2 Workplace is prepared as per job requirement. 1.3 <u>Tools, equipment</u> and drawing are collected as per job requirement. 1.4 <u>Elements</u> for preparing electrical circuits are identified and collected 1.5 <u>Electrical circuit diagram</u> is understood and analyzed. 1.6 <u>Electrical circuit</u> wirings are carried out.
2. Install one lamp control by SPST and SPDT switch.	2.1 Drawing and specification sheet is collected 2.2 Tools and materials are collected as per job requirement 2.3 Working board is marked as per lay out. 2.4 SPST and SPDT switch are installed and connected 2.5 Connection is checked and tested.
3. Install one calling bell control from two point.	3.1 Drawing and specification sheet is collected 3.2 Tools and materials are collected as per job requirement 3.3 Working board is marked as per lay out. 3.4 Calling bell is installed and connected as per drawing 3.5 Connection is checked and tested.
4. Prepare the circuit of LED Lamp.	4.1 Drawing and specification sheet is collected 4.2 Tools and materials are collected as per job requirement 4.3 Working board is marked as per lay out. 4.4 LED Lamp is connected and tested
5. Install ceiling fan with regulator.	5.1 Drawing and specification sheet is collected 5.2 Tools and materials are collected as per job requirement 5.3 Working board is marked as per lay out. 5.4 Ceiling Fan with regulator is installed and connected as

	per drawing 5.5 Connection is checked and tested.
6. Install power socket.	6.1 Requirement of power socket is identified. 6.2 Tools are collected as per job requirement 6.3 Wall is marked as per lay out. 6.4 Power socket is installed with earthing line. 6.5 Connection is checked and tested.

Range of Variables

Variable	Range May Include but not limited to:
1. Tools and equipment	1.1 Tools: 1.1.1. Measuring tape (30m) 1.1.2. Tri- square 1.1.3. Wire stripper 1.1.4. Adjustable wrench 1.1.5. Files: (flat, round, half round) 1.1.6. Hand hacksaw 1.1.7. Hammers: (ball peen, claw) 1.1.8. Pliers (combination pliers, cutting pliers, diagonal cutting pliers, long nose pliers) 1.1.9. Punches 1.1.10. Screwdrivers (star, negative, positive) 1.1.11. Electrician knife 1.2 Equipment: 1.2.1 Multi meter/AVO meter
2. Elements	2.1 Switch 2.2 Socket 2.3 Cable 2.4 Resistive load 2.5 Inductive load
3. Electrical circuit diagram	3.1 Series circuit 3.2 Parallel circuit 3.3 Series-parallel circuit 3.4 Lamp circuits
4. Electrical circuit	4.1 Series circuits 4.2 Parallel circuits 4.3 Series-parallel circuits 4.4 One Lamp controlled from one point 4.5 One Lamp controlled from two points 4.6 Two Lamps controlled from two points 4.7 Two Lamps controlled from three points

	4.8 Tube Light Connection 4.9 Power supply circuit
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Curricular Content Guide:

1. Underpinning Knowledge	1.1 Recognize an electrical circuit and components. 1.2 Understand electrical circuit diagram. 1.3 Types of circuits used in residential building. 1.4 Diagrams of SPST and SPDT wiring.
2. Underpinning Skills	2.1 Carrying out electrical circuit wirings. 2.2 Installing and connected SPST and SPDT switch. 2.3 Installing and connected calling bell drawing. 2.4 Installing Power socket with earthing line.
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Various kinds of electrical/electronic components, tools and equipment 4.3 Papers 4.4 Work books

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 identified elements for preparing electrical circuits. 1.2 analyzed electrical circuit diagram understood. 1.3 marked working board as per lay out. 1.4 installed power socket with earthing line. 1.5 checked and tested connection.
2. Methods of Assessment	Methods of assessment may include but not limited to: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done at a training center.

	3.2 Assessment should be done by the certified assessor of TÜV SÜD.
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Unit of Competency: PERFORM WIRE OR CABLE TERMINATION	Nominal Duration: 5 hrs.	Unit Code: SICIP-LE-BE-03-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to perform wire or cable termination. It specifically includes the tasks of collecting tools, equipment and materials, identifying cables lugs, connecting cables and maintaining workplace, tools and materials.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Collect tools, equipment and materials.	1.1 Personal protective equipment (PPE) is identified and used. 1.2 <u>Appropriate tools and equipment</u> are identified and collected based on the termination requirements. 1.3 <u>Materials</u> of termination works are identified and arranged as per termination requirements.
2. Identify cables lugs.	2.1 Specifications and manufacturer manuals related to cable lugs are reviewed. 2.2 <u>Types of cable lugs</u> are identified. 2.3 <u>Cable lug sizes</u> are identified to ensure compatibility with cables and electrical requirements. 2.4 Compatibility of cable lugs and sizes is confirmed based on specifications.
3. Connect cables (1-4 mm ²).	3.1 Cables sized 1-4 mm ² are measured, cut to length, and insulation is stripped. 3.2 Cable ends are cleaned, and conductors are twisted or tinned if required. 3.3 Appropriate connectors or terminals compatible with 1-4 mm ² cables are selected. 3.4 Cables are connected to terminals or connectors using crimping. 3.5 All connections are insulated with heat shrink tubing, or electrical tape. 3.6 Electrical continuity is tested using appropriate testing

	instruments.
4. Maintain workplace, tools and materials.	4.1 Workplace is cleaned and organized to ensure safety and efficiency. 4.2 Tools are inspected regularly for damage or wear and maintained according to manufacturer guidelines. 4.3 Tools and equipment are stored in designated areas following safety and organizational procedures.

Range of Variables

Variable	Range May Include but not limited to:
1. Appropriate tools and equipment	1.1 Wire cutter 1.2 Wire stripper 1.3 Knife 1.4 Crimping tools
2. Materials	2.1 Cables 2.2 Connectors/lugs 2.3 Terminals 2.4 Insulating materials
3. Types of cable lugs	3.1 Ring/O Lugs 3.2 Fork/U Lugs 3.3 Pin/I Lugs
4. Cable lug sizes	4.1 1.5mm ² 4.2 2.5mm ² 4.3 4.0mm ²

Curricular Content Guide:

1. Underpinning Knowledge	1.1 Identify appropriate tools and equipment for the termination requirements. 1.2 Recognize wire or cable termination. 1.3 Identify cables lug's, purpose and types. 1.4 Sizes of cable lug. 1.5 Methods of cables connection.
2. Underpinning Skills	2.1 Measuring and striped the right length of insulation. 2.2 Connecting cables to terminals or connectors using crimping. 2.3 Applying insulation tape or heat shrink tubing.
3. Underpinning	3.1 Commitment to occupational health and safety

Attitudes	3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Required tools, equipment and materials. 4.3 Work books 4.4 Learning manuals

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 identified appropriate tools and equipment and materials. 1.2 identified types of cable lugs and sizes. 1.3 confirmed cable lugs and sizes based on specifications. 1.4 connected cables to terminals or connectors using crimping. 1.5 tested electrical continuity using appropriate testing instruments.
2. Methods of Assessment	Methods of assessment may include but not limited to: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done at a training center. 3.2 Assessment should be done by the certified assessor of TÜV SÜD.

Unit of Competency: PERFORM CABLE GLANDING	Nominal Duration: 5 hrs.	Unit Code: SICIP-LE-BE-04-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to perform cable glanding. It specifically includes the tasks of collecting tools, equipment and materials, identifying		

cables gland and connecting plastic gland (PG-gland).

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Collect tools, equipment and materials.	1.1 Personal protective equipment (PPE) is identified and used. 1.2 <u>Appropriate tools</u> are identified and collected based on the glanding requirements. 1.3 <u>Materials</u> of glanding works are identified and arranged as per glanding requirements.
2. Identify cables gland.	2.1 Technical manuals, datasheets, and relevant standards are understood. 2.2 <u>Types of cable glands</u> are identified. 2.3 Materials used in cable glands and their specific applications are understood. 2.4 Cable gland sizes and thread types are identified against cable and conduit specifications.
3. Connect plastic gland (PG-gland).	3.1 Relevant PG-gland specifications and installation instructions are understood. 3.2 Appropriate PG-gland size is selected based on the cable diameter and type. 3.3 Cable is cut and stripped as per measurement. 3.4 <u>PG-gland components</u> are assembled on the cable in correct order. 3.5 PG-gland is tightened using appropriate tools. 3.6 Workplace and tools are cleaned and maintained. 3.7 Tools and equipment are stored in designated areas.

Range of Variables

Variable	Range
	May Include but not limited to:
1. Appropriate tools	1.1 Cutters 1.2 Knife 1.3 Wrench/Spanner 1.4 Measuring tape
2. Materials	2.1 Cables 2.2 Glands

	2.3 Insulating materials 2.4 Marker pen
3. Types of cable glands	3.1 Metallic 3.2 Non-metallic 3.3 PG-gland 3.4 Explosion-proof 3.5 Armored 3.6 Unarmored
4. PG-gland components	4.1 Main Body 4.2 Clamping Ring/Claw 4.3 Seal 4.4 Locknut/Sealing Nut

Curricular Content Guide:

1. Underpinning Knowledge	1.1 Identify appropriate tools based on the glanding requirements. 1.2 Glanding works materials. 1.3 Types of cable glands and specific application. 1.4 Define cable gland sizes and thread types. 1.5 Understand PG-gland components. 1.6 Tools and equipment storing procedure.
2. Underpinning Skills	2.1 Collecting appropriate tools based on the glanding requirements. 2.2 Selecting appropriate PG-gland size based on the cable diameter and type. 2.3 Assembling PG-gland components on the cable in correct order. 2.4 Tightening PG-gland using appropriate tools. 2.5 Cleaning technique of workplace and tools.
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Required tools and materials for the job

	4.3 Manuals 4.4 Work books
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Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 identified and collected appropriate tools based on the glanding requirements. 1.2 understood technical manuals, datasheets, and relevant standards. 1.3 understood relevant PG-gland specifications and installation instructions.
2. Methods of Assessment	Methods of assessment may include but not limited to: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done at a training center. 3.2 Assessment should be done by the certified assessor of TÜV SÜD.

Unit of Competency: PERFORM CHANNEL WIRING	Nominal Duration: 35 hrs.	Unit Code: SICIP-LE-EIM-05-O
Unit Descriptor: This unit of competency requires knowledge, skills and attitude to perform channel wiring. It specifically includes the tasks of preparing for work, carrying out channel wiring, installing boards and other accessories, testing the wiring and maintaining and store tools and materials.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are described in the range of variables).

Elements of Competency	Performance Criteria
1. Prepare for work.	1.1 Personal Protective Equipment (PPE) is collected and worn as per job requirement. 1.2 Workplace is prepared as per job requirement. 1.3 <u>Tools</u> and <u>materials</u> are selected and collected as per job requirement.
2. Carry out channel wiring.	2.1 Wiring layout is drawn according to supplied drawing. 2.2 Rowel plug points are located, drilled and inserted

	as per procedure. 2.3 Bottom parts of the channels are installed and screwed. 2.4 <u>Boards and different fixtures</u> are collected, fitted and connected as per diagram. 2.5 Cables are laid as per specification. 2.6 The bottom parts of the channels are covered with upper part of the channel
3. Install boards and other accessories.	3.1 Circuit materials are placed on the board. 3.2 Circuit accessories are fitted and connected using wire
4. Test the wiring.	4.1 Electrical circuit connection is checked using meter. 4.2 Circuit is <u>tested</u> with connected power supply.
5. Maintain and Store tools and materials.	5.1 Electrical tools/instruments are checked for proper operation. 5.2 Electrical tools/instruments are maintained in accordance to manufacturer's specification. 5.3 Electrical tools/instruments stored in accordance to workplace procedures/policy.

Range of Variables

Variable	Range (Includes but not limited to:)
1. Tools and Materials	1.1 T-Square 1.2 Sprit level 1.3 Channel 1.4 PVC box 1.5 Insulation taps 1.6 Wire 1.7 Screw 1.8 Switch board (different type) 1.9 Switch 1.10 Socket 1.11 Fuse
2. Boards and different fixtures	2.1 Boards: 2.1.1 Plastic board 2.2 Different Fixtures: 2.2.1 Switches 2.2.2 Sockets 2.2.3 Fan regulator 2.2.4 Fuse 2.2.5 Ceiling rose 2.2.6 Indicating lamp 2.2.7 Lamp Holders

	2.2.8 LED Lights 2.2.9 LED Tube Lights 2.2.10 Circuit Breakers (CB)
3 Test	3.2.1 Insulation resistance test 3.2.2 Continuity test 3.2.3 Polarity test

Curricular Content Guide

1. Underpinning Knowledge	1.1 Function and types of board and different fixtures. 1.2 Wiring circuits and their application 1.3 Connection and connecting point with the load. 1.4 Channel wiring and testing methods 1.5 Circuit wiring, installation and maintenance.
2. Underpinning Skills	2.1 Using Personal Protective Equipment (PPE). 2.2 Using tools, equipment and materials. 2.3 Selecting appropriate tools and equipment. 2.4 Selecting appropriate materials. 2.5 Drawing wiring lay out. 2.6 Preparing basic electrical circuit. 2.7 Locating points and installing and screwing channel. 2.8 Fitting boards and different fixtures. 2.9 Performing channel wiring and testing. 2.10 Maintaining electrical tools/instruments.
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, sub-ordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual). 4.2 Various kinds of electrical components, tools and equipment. 4.3 Layout and diagram. 4.4 Workplace manual. 4.5 Work books.

Assessment Evidence Guide

1. Critical Aspects of	Assessment required evidence that the candidate:
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Competency	1.1 drawn wiring layout according to supplied drawing. 1.2 located points for rowel plug 1.3 installed rowel plugs and channels 1.4 fitted and connected boards and different fixtures as per diagram. 1.5 laid cables. 1.6 tested circuit with connected power supply. 1.7 checked electrical tools/instruments for proper operation.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written Test 2.2 Practical Demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done at a training center. 3.2 Assessment should be done by the certified assessor of TÜV SÜD.

Unit of Competency: PERFORM CONDUIT WIRING	Nominal Duration: 40 hrs.	Unit Code: SICIP-LE-BE-06-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to perform conduit wiring. It specifically includes the tasks of collecting tools, equipment and materials, installing conduits and lay cables, installing boards and other accessories of wiring, test the wiring and maintaining workplace, tools and materials.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Collect tools, equipment and materials.	1.1 Personal Protective Equipment (PPE) is collected and worn as per job requirement. 1.2 Workplace is prepared as per job requirement. 1.3 <u>Tools</u> and <u>materials</u> are selected and collected as per job requirement.
2. Install conduits and lay cables.	1.1 Wall is marked and cut the groove as per layout diagram. 1.2 Conduit is installed inside the groove using saddle clamp.

	1.3 Switch box and junction box are installed inside the groove using clipping nail. 1.4 Cable is laid into the conduit from switch board to load point with the help of fish / GI wire as required. 1.5 Electrical fittings are installed on the wall as per electrical diagram. 1.6 Cable is connected with fittings 1.7 Cable is laid from switch board to SDB and SDB to MDB 1.8 All Connections are checked using meter. 1.9 Circuit is operated by power supply
3. Install boards and other accessories of wiring.	3.1 Switch board, circuit breakers, energy meter, busbar and 3.2 connectors /insulator are installed. 3.3 Required fitting materials are placed on the board. 3.4 Switch, socket and Accessories are fitted and connected using wire.
4. Test the wiring.	4.1 Electrical circuit connection is checked using meter. 4.2 Circuit is tested with connected power supply.
5. Maintain workplace, tools and materials.	5.1 Work area is cleaned in accordance with workplace procedures. 5.2 Unused materials are stored for re-use or disposed following workplace procedures. 5.3 Waste and scrap materials are disposed with following workplace and environmental procedures. 5.4 Inventory of tools equipment are conducted and recorded as per checklist. 5.5 Tools and equipment are cleaned and stored as per manufacturer's recommendation in appropriate location.

Range of Variables

Variable	Range May Include but not limited to:
1. Tools	1.1 Wire stripper 1.2 Mallet 1.3 Cold Chisels 1.4 Hacksaw 1.5 Hammers: (a) ball peen, 1.6 Hand drill machine 1.7 Measuring tape

	1.8 Electrician knife 1.9 Pliers: (a) combination pliers, (b) side cutting pliers, (c) diagonal cutting pliers, (d) long nose pliers 1.10 Plumb bob 1.11 Screw drivers:(a) star, (b) flat, (c) connecting 1.12 Try square 1.13 Neon tester 1.14 Ladder 1.15 Sprit level 1.16 Megger/ Insulation tester 1.17 Multimeter / AVO Meter 1.18 Electric drill machine 1.19 Electric grinding machine
2. Materials	2.1 RM cable 1.5 2.2 RM cable 2.5 2.3 RM cable 4.0
3. Switch, socket	3.1 One-way switch 3.2 Two-way switch 3.3 Intermediate switch 3.4 Dimmer switch 3.5 3-pin socket (13A) 3.6 2-pin socket (6A) 3.7 Universal socket
4. Accessories	4.1 Bulb holders, 4.2 Ceiling rose 4.3 Fuse 4.4 PVC conduits 4.5 Metal conduits 4.6 Bends, elbows, couplers 4.7 Switch Boards 4.8 Cable Clips and Clamps 4.9 Wires and Cables 4.10 Junction Boxes 4.11 Indicator Lamps 4.12 Bell Push / Doorbell Switch
5. Test	5.1 Insulation resistance test 5.2 Continuity test 5.3 Polarity test

Curricular Content Guide:

1. Underpinning Knowledge	1.1 Understand layout diagram. 1.2 Types and function of switches, sockets, ceiling rose and fittings. 1.3 Different types of fittings.
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	1.4 Fittings and fixture installation. 1.5 Cable laying procedure. 1.6 Electrical power supply system. 1.7 Electrical circuit test procedures. 1.8 Materials stored for re-use or disposed procedures.
2. Underpinning Skills	2.1 Selecting appropriate tools, materials and fittings related to the installing fittings. 2.2 Using tools and fittings related to the installing fittings and fixtures. 2.3 Checking layout diagram. 2.4 Marking and cutting the groove on the wall and into the wall. 2.5 Installing fittings and fixture as per drawing. 2.6 Connecting cable with load point and switch board as per diagram. 2.7 Checking and testing circuit. 2.8 Maintaining workplace, tools and materials.
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety. 3.2 Promptness in carrying out activities. 3.3 Sincerity and honesty to duties. 3.4 Environmental concerns. 3.5 Eagerness to learn. 3.6 Tidiness and timeliness. 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace.
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Various kinds of electrical components, tools and equipment and materials. 4.3 Layout and diagram. 4.4 Workplace manual. 4.5 Work books.

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 selected and collected tools and materials. 1.2 marked and cut the wall as per layout diagram. 1.3 installed conduit inside the groove using saddle clamp. 1.4 installed electrical fittings and fixtures on the wall. 1.5 checked connection the cable with the load.
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	1.6 tested circuit with connected power supply.
2. Methods of Assessment	Methods of assessment may include but not limited to: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done at a training center. 3.2 Assessment should be done by the certified assessor of TÜV SÜD.

Workshop/Lab Facility Standard

Course Name:	Building Electrification
Number of Trainees:	20

Course-wise Training Space (Theoretical Classroom, Workshop/Lab/Classroom cum Workshop):

- Classroom – 400 sft
- Workshop/ lab – 600 sft

Major Training Equipment and Training Facilities:

Sl. No.	Major Equipment and Training facilities	Required facilities
1.	Computer/Laptop with internet facility	1
2.	Multimedia projector & screen	1
3.	Wiring Board	20
4.	Main Distribution Board with busbar, CB and other accessories (Training model)	1
5.	Sub Distribution Board with busbar, CB and other accessories (Training model)	10
6.	Magnetic contactor (coil voltage 220V and 400V)	5
7.	Push switch	30
8.	Generator	1
9.	Electric drill machine	5
10.	Electric grinding machine	2
11.	Soldering Stations	5
12.	Energy Meter (1 Phase)	10
13.	Energy Meter (3 Phase)	1
14.	Multi meter/AVO meter	10
15.	Earth tester	1
16.	Insulation tester	2
17.	Clamp Meter	2
18.	Phase sequence meter	1
19.	Single phase induction motor	2
20.	Ceiling fan	10

The following conditions must be fulfilled –

- The institute shall not use the same facilities for any other projects/organizations offering a similar course.
- The institute must provide sufficient evidence to prove ownership of the proposed training equipment.

The list denotes the minimum training equipment and facility required to effectively conduct training for a specific course. Additionally, the institute must ensure that all other necessary

training tools, equipment, and furniture are available to meet the requirement of competency standards (CS) provided by SICIP.

For the operation of training course on **Building Electrification** the institute must ensure the availability of at least 80% of the major training equipment and training facilities (according to the CS) to be eligible for SICIP training delivery. If the score is below 80%, the remaining equipment and facilities need to be installed before the commencement of the training.

The institute will also provide all other hand tools and power tools as per CS for 20 trainees. Also, they will arrange adequate seating arrangement and classroom setup for the 20 trainees.